Day 4: mathematical diagnostic

Name: _____ Date: _____

60 minutes total. No calculator, notes, internet or AI on the independent pass. Show your reasoning. If stuck, leave the useful work you have and move on. This is not a speed contest.

3 minutes setup · up to 30 minutes core · up to 20 minutes for two selected follow-ups · 7 minutes for your log. The tutor will supply follow-ups. Do not work through the whole reserve book.

If you finish a block early, check your work or continue to the next block with the tutor. Do not exceed the overall limit.

F1-3

Let $f(x)=\sqrt{(x-1)}$ and $g(x)=3-x$. Find the real domain of $f(g(x))$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

F2-3

A function on R satisfies $f(x)+2f(1-x)=3x$ for every real x. Find $f(0)$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A4-3

Find all real x satisfying $x^4 - 3x^2 + 2=0$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

F1-2

Let $f(x)=x^2+1$ and $g(x)=2x-3$ for all real x. Find $f(g(2))$ and $g(f(2))$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

F2-2

For all real x,y, $f(x+y)=f(x)+f(y)$, and $f(1)=3$. Find $f(2)$, $f(-1)$ and $f(1/2)$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A4-2

For $x \neq 0$, $x+1/x=3$. Find x^2+1/x^2.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

F2-1

A real-valued function on R satisfies f(x+y)=f(x)+f(y). What do the substitutions x=y=0 and y=−x tell you about f(0) and f(−x)?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A5-1

For real logarithms to base b, state the restrictions on b and on the argument. State log_b(uv) for positive u,v.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

Session log

Which question was hardest to start, and what did you try?

_____

Was any question familiar, hinted, interrupted, or affected by outside study? Record IDs and details.

_____

Follow-up IDs attempted: _____ / _____